GAURAV PAGARE

Software Engineer – ML Systems & Infrastructure

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SUMMARY

Founding engineer of an AI dubbing company's technical team. Built its speech-recognition training platform from an empty repository to distributed multi-GPU training running unattended on cloud, and wrote the founding commit of the dubbing product that is now the company's live application. Comfortable end to end — data pipelines, DeepSpeed/NCCL debugging, Prefect/MLflow orchestration, model and provider evaluation, and the product surface on top. Now reports directly to the CEO.

EXPERIENCE

Ant Systemz Pvt. Ltd. — Software Engineer - ML Systems & Infrastructure

Apr 2025 – Present

Mumbai · First engineering hire; joined the day the technical team was formed and grew with it to ~15 engineers · 500+ commits across 10 repositories.

Speech recognition training platform (Whisper fine-tuning · 195 commits)

- Built an end-to-end ASR training platform from scratch — audio preprocessing (VAD via pyannote/Silero/Kaldi, Demucs source separation, segment-level deduplication, augmentation, QC gates) through distributed training, checkpoint export and cloud sync.
- Productionized multi-GPU training on **DeepSpeed ZeRO-2** across 4-GPU nodes; diagnosed and fixed NCCL collective failures, torchrun TCPStore rendezvous hangs, per-rank CUDA device binding, distributed LR-schedule desync, and multi-GPU finalize deadlocks on GCP Blackwell instances.
- Orchestrated preprocessing and training with **Prefect + MLflow**; added GCS autoresume, hardware-aware GPU/CPU launch, a Docker Compose training operator, and pre-training readiness checks — enabling unattended multi-day runs.
- Containerized the stack for **GCP and AWS**, with regional Cloud Build push, CI smoke tests, and dual-cloud config parity, and then continued training on **Vertex AI**.
- Versioned the data pipeline on GCS with incremental splits, race-condition-safe shared locks, and processed-version detection.
- Fine-tuned Whisper with **PEFT/LoRA**; fixed Hugging Face checkpoint export against live PEFT adapters and hardened decoder-label and validation contracts.
- Built the pipelines to operate over a corpus of **2,000 hours per language across 20 languages**.

VOIT — the company's production dubbing platform (104 commits · founding commit Apr 2025)

- **Wrote the first commit of the product in April 2025** and built the original version end to end — transcription, translation and synthesis, Devanagari script handling, S3-backed staged storage with thread parallelism, AWS Lambda for audio extraction and auth, and the migration to PostgreSQL.
- Lead engineer as it became the company's live application, adding user workspaces and projects and planning the MySQL→PostgreSQL consolidation.
- Shipped a **dual-provider Voice Library** with emotion variants, per-line voice variants and self-healing repair; normalized voice enrolment so a single enrolment works across providers with different format and size ceilings — trimming uploads to each provider's limit rather than rejecting them.
- Eliminated synthesis hallucination on long input through **chunked TTS**; added duration capping and speech-preserving timed placement so dubbed audio fits the original timing without pitch damage.
- Made batch translation resilient to partial model output, and skipped redundant synthesis calls on same-language jobs.
- **Cut editor page load from 4,523 MB to 38 MB transferred (118×)** and from 13 GB to 176 MB of decoded resources, measured on a 7-minute 11-speaker multitrack job — by caching in-flight promises rather than resolved buffers, serving compressed previews instead of raw WAV stems, lazy-rendering waveforms via IntersectionObserver, and clamping a clip canvas being sized to 11,373,603 px (a ~3 GB allocation per clip, past the browser's limit).
- Diagnosed a long-standing playback bug that three previous fixes had missed, by tracing which layer was actually audible rather than the one that appeared broken — the real cause was a dead code path that had been failing silently for months.

AutoDub — separate internal testing platform (19 commits)

- Built and maintain AutoDub, an internal tool distinct from the production product and not customer-facing: a place to evaluate new speech approaches against our own use cases, and where a few team members run interim dubbing work.
- Benchmarked commercial and open-source speech and translation providers against the company's language mix, across regional accents, then designed a **dual-provider abstraction with fallback** so a capability gap in any one vendor never blocks work.
- Prototyped an automated **dub QC** step that grades accent, dialect and synthesis quality with an audio model, replacing manual listening passes.

Cloud, data and storage infrastructure

- Built multi-worker Google Drive → **S3 / Wasabi** migration and metadata-extraction services with threaded parallel transfer, S3 access points, and purpose-built IAM users, roles, buckets and policies; ran and monitored the production migration.
- Evaluated Wasabi against AWS for the real workload and recommended AWS-only — removing a redundant storage tier and its recurring cost.
- Produced corrected infrastructure cost sheets and a layered technical architecture document for external consultant review.

Dub-Storage — enterprise media platform (Aug–Dec 2025 · 97 commits)

- Built a role-based media platform — folder hierarchies, cut-paste, restore, upload/download tracking, activity reporting and rate limiting; resolved database connection-pool exhaustion under load and enabled large-file uploads.

Technical leadership and ownership

- Onboarded and trained every engineer hired after me — all first-job hires — as the team grew to ~10.
- Served as the team's primary technical point of contact: task breakdown, sprint planning, delivery follow-through and escalation. Reported directly to the CTO; **now reports directly to the CEO** with a weekly engineering report on delivery, blockers and decisions.
- Owned architecture and build-vs-buy decisions across storage, model providers, GPU/instance sizing and orchestration tooling.
- Led vendor relationships and commercial negotiations across cloud and AI suppliers — pricing, credits and integration escalations.

Whole New Level — Frontend Web Developer

Aug 2023 – Mar 2025

Mumbai

- Built and maintained 20+ WordPress sites, including the Communities Foundation of Texas (CFT) and CCRSM projects; implemented WooCommerce flows, site migrations and performance optimization.

Ahen Pvt. Ltd. — Frontend Developer Intern

Dec 2022 – Apr 2023

Pune

- Built responsive React interfaces to pixel-perfect designs within a function-based architecture.

SKILLS

ML & Speech PyTorch · DeepSpeed (ZeRO) · Hugging Face Transformers · PEFT/LoRA · Whisper · VAD (pyannote, Silero, Kaldi) · Demucs · speaker diarization & clustering (AHC, spectral) · TTS/STT · voice cloning · data augmentation

MLOps Prefect · MLflow · Docker / Docker Compose · torchrun & NCCL · multi-GPU distributed training · autoresume & checkpointing · CI

Cloud GCP (Vertex AI, GCS, Cloud Build, GCE) · AWS (EC2, S3, Lambda, IAM, access points) · Akamai · Wasabi · rclone

Backend Python · FastAPI · Flask · Node.js · PostgreSQL · MySQL · REST APIs

Frontend React · TypeScript · JavaScript · Next.js

EDUCATION & PUBLICATIONS

AISSMS College of Engineering, Pune — B.E. Computer Engineering, 8.85 CGPA

2020 – 2024

Undergraduate research: real-time Indian Sign Language recognition — custom Marathi dataset, MediaPipe keypoint extraction, BiLSTM sequence model.

- Indian Sign Language Recognition Using Holistic Body Pose Estimation and Deep Learning — Journal of Technology, May 2024
- Indian Sign Language Recognition Using CNN Inception V3: A Review — IJRASET, Feb 2024